Fiscal Year:	FY 2023	**Task Last Updated:**	FY 12/27/2022
PI Name:	Narayanan, Anand Ph.D.		
Project Title:	Simulated Microgravity-Induced Systemic Inflammation and Its Impact on Circulatory Function and Structure		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	anarayan09@gmail.com	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	203 785 6055
Organization Name:	Florida State University		
PI Address 1:	College of Human Sciences		
PI Address 2:	242 Sandels Building, 120 Convocation Way		
PI Web Page:			
City:	Tallahassee	**State:**	FL
Zip Code:	32306-0001	**Congressional District:**	2
Comments:	Former email address: anarayanan@fsu.edu		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	03/01/2019	**End Date:**	07/31/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	7
No. of Bachelor's Candidates:	24	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2024 per NSSC information (Ed., 11/13/23) NOTE: End date changed to 7/31/2023 per NSSC information (Ed., 1/4/23) NOTE: End date changed to 5/31/2022 per F. Hernandez/ARC and NSSC information (Ed., 6/16/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Delp, Michael Ph.D. (MENTOR: Florida State University)		
Grant/Contract No.:	80NSSC19K0426		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP Long-term exposure to the space environment poses potential harm to astronaut health, including elevated morbidity and mortality risk for cardiovascular disease and heightened immune dysregulation. The cardiovascular system has adapted to gravity on Earth. The removal of hydrostatic pressure gradients during exposure to weightlessness has been thought to be the primary initiator of systemic circulatory adaptations during spaceflight. The circulation transports blood and lymph, which contain immune factors. Imbalances of immune factors can lead to increased susceptibility to infections and/or inflammatory pathologies, as well as affect the cardiovascular system. However, there is a paucity of data investigating the interplay between the cardiovascular and immune systems. This fellowship project is tied with NASA project grant 80NSSSC19K1322 (Principal Investigator Michael Delp), which has been delayed with the primary experimental effort just having completed in December 2020. Given the original project delay, I have been investigating the effects of spaceflight on the cerebral and heart circulation from Rodent-Research 9 mission (NNX16AC28G; PI Michael Delp). I am also continuing this line of research, i.e., investigating multiple organ bed vasculature adaptations to spaceflight, through a tissue sharing opportunity via International Space Station (ISS) National Lab via Rodent Reference Mission-2 (RRRM2), studying the cerebral, coronary, GI (gastrointestinal), and hindlimb vascular tissues in response to spaceflight, old age, and long-term recovery. Given delays, I have adapted the original proposal to the following aims based on the unique opportunities available to study actual spaceflight samples, in contrast to the original proposal's objectives to study simulated weightlessness adaptations. The new proposed aims include: Aim 1: To determine whether spaceflight alters the cerebral and coronary vasculature. Aim 2: To determine whether spaceflight exposure leads to adaptations in the cerebral, coronary, GI, and hindlimb vasculature in response to the single and combined effects of old age +/- recovery. Aim 3: To determine the long-term single and combined effects of simulated microgravity and/or radiation on GI and hindlimb vasculature.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In completing these studies, I have shown how the vascular state from multiple organ beds adapts to various spaceflight conditions (actual spaceflight, simulated microgravity and/or radiation, etc.). These studies will enhance our understanding of space biological adaptations of the cardiovascular system, and furthermore, increase our understanding of medical conditions astronauts develop while in space. In studying spaceflight biomedical adaptations, we also increase our knowledge of Earth-based biomedical science by investigating novel pathways activated by the unique spaceflight environment (i.e. gravity, low-dose radiation, isolation, etc.) that may provide new perspectives of how medical conditions develop and/or biology evolve on Earth. These studies provide foundational work that I will further characterize through various on-going and future projects.
Task Progress:	Project Description: Long-term exposure to the space environment poses potential harm to astronaut health, including elevated morbidity and mortality risk for cardiovascular disease and heightened immune dysregulation. The cardiovascular system has adapted to gravity on Earth. The removal of hydrostatic pressure gradients during exposure to weightlessness has been thought to be the primary initiator of systemic circulatory adaptations during spaceflight. The circulation transports blood and lymph, which contain immune factors. Imbalances of immune factors can lead to increased susceptibility to infections and/or inflammatory pathologies, as well as affect the cardiovascular system. However, there is a paucity of data investigating the interplay between the cardiovascular and immune systems. This fellowship project is tied with NASA project grant #80NSSSC19K1322 (Principal Investigator Michael Delp), which had been delayed with the primary experimental effort having been completed in December 2020. Given the original project delay, I had been investigating the effects of spaceflight on the cerebral and heart circulation from Rodent-Research 9 mission (NNX16AC28G; Principal Investigator Michael Delp). I am also continuing this line of research, i.e., investigating multiple organ bed vasculature adaptations to spaceflight, through tissue sharing opportunities via the International Space Station (ISS) National Laboratory via Rodent Research Reference Mission-2 (RRRM2), studying the cerebral, coronary, gastrointestinal (GI), and hindlimb vascular tissues in response to spaceflight, old age, and long-term recovery. Given delays, I have adapted the original proposal to the following aims based on the unique opportunities available to study actual spaceflight samples, in contrast to the original proposal's objectives to study simulated weightlessness adaptations. The new proposed aims included: Aim 1: To determine whether spaceflight alters the cerebral and coronary vasculature. Aim 2: To determine whether spaceflight exposure leads to adaptations in the cerebral, coronary, GI, and hindlimb vasculature in response to the single and combined effects of old age +/- recovery. Aim 3: To determine the long-term single and combined effects of simulated microgravity and/or radiation on GI and hindlimb vasculature. Project Status: The Post-Doctoral Fellowship project was tied with NASA Grant #80NSSC18K0808: Radiation, Simulated Weightlessness and Countermeasures: Effects on Cerebral and Coronary Vascular Function and Structure. This project (80NSSC19K1322) was delayed and was not completed until Fall 2020. Thus, the experimental work tied to this Post-Doctoral Fellowship had been adapted for studies from Rodent Research-9 (NNX16AC28G), ongoing studies from Rodent Reference Mission-2 via ISS National Lab, and the more recent NASA Project 80NSSC18K0808. Aim 1 has been completed, manuscripts in preparation. Aim 2 is ongoing, with 75% of experimental work completed (tissue harvest and processing), anticipating completion and submission of manuscripts, annual report, etc., by Summer 2023. Aim 3 is ongoing, with 75% of experimental work completed, anticipating completion and submission of manuscripts, annual report, etc., by Fall 2023.

Bibliography Type:	Description: (Last Updated: 05/22/2024)
Abstracts for Journals and Proceedings	Narayanan SA, Vasserman D, Linarte M, Mitchell R, Mao X, Willey J, Tarasova OA, Bagher P, Zawieja DC, Delp MD. "Spaceflight induces protein expression alterations related to mouse basilar artery function, oxidative stress state, and DNA damage." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021
Abstracts for Journals and Proceedings	Jorrin R, Vasserman D, Narayanan SA, Delp M. "Long-term effects of simulated spaceflight exposure assessed by Doppler ultrasonography." Florida Undergraduate Research Conference, Orlando, Florida, February 18-19, 2022. Abstracts. Florida Undergraduate Research Conference, Orlando, Florida, February 18-19, 2022. , Feb-2022
Abstracts for Journals and Proceedings	Zuniga D, Neustadter H, Liberato J, Narayanan SA, Delp M. "Long-term effects of simulated spaceflight exposure on the coronary artery." Florida Undergraduate Research Conference, Orlando, Florida, February 18-19, 2022. Abstracts. Florida Undergraduate Research Conference, Orlando, Florida, February 18-19, 2022. , Feb-2022
Articles in Peer-reviewed Journals	Patrick G, Leonard M, Flohr M, Sangha N, Torres AA, Hartenstein B, Reyes-Ortiz C, Narayanan SA, Chang M, Fleischer C. "Prevalence of self-reported dermatologic symptoms among farmers living in Gracias a Dios, Honduras." JAAD International. 2022 Sep 1;8:89-91. Research Letter. https://doi.org/10.1016/j.jdin.2022.05.012 , Sep-2022
Articles in Peer-reviewed Journals	Jamison DA Jr, Anand Narayanan S, Trovão NS, Guarnieri JW, Topper MJ, Moraes-Vieira PM, Zaksas V, Singh KK, Wurtele ES, Beheshti A. "A comprehensive SARS-CoV-2 and COVID-19 review, Part 1: Intracellular overdrive for SARS-CoV-2 infection." Eur J Hum Genet. 2022 May 16;30:889-98. Review. https://doi.org/10.1038/s41431-022-01108-8 ; PMID: 35577935; PMCID: PMC9108708 , May-2022